PhysAlign: Physics-Coherent Video Generation through Feature and 3D Representation Alignment

Supplementary Material

Table 1. User study results. We evaluate the user's preference from three aspects: Motion Physics Quality, Identity Physics Quality and Overall Video Quality. Higher is better.

	CogVideoX	Hunyuan	Wan2.2	Ours
Motion Physics Quality	1.671	1.543	2.557	**3.236**
Identity Physics Quality	1.757	1.679	2.536	**3.343**
Overall Video Quality	1.771	1.607	2.643	**3.286**

Table 3. Additional ablation Physical Invariance Score(PIS) on WISA Test Set. Higher is better.

PIS	Train on WISA	**Ours**
a_x	0.538	**0.592**
a_y	0.521	**0.598**
v_x	0.753	**0.764**
v_y	0.674	**0.725**
Δl	0.399	**0.436**

A. User Study

We conduct a user study to compare our method against baseline approaches, as summarized in Tab. 1. Participants evaluate the generated videos from three perspectives: (1) *Motion Physics Quality*, which measures how well the object motion adheres to physical laws; (2) *Identity Physics Quality*, which reflects whether the subject's deformation is consistent with physics; and (3) *Overall Video Quality*, which captures general perceptual quality including visual fidelity and temporal consistency.

Each participant rates every video on a scale from 1 (poor) to 4 (best), and we report the average score for each method. The results show that our approach consistently receives higher human preference across all three aspects, demonstrating the effectiveness of the proposed physics-aware generation framework. We show the visualization of our user-study page in Fig. 1.

B. Experiment Details

We train our model on 4×H100 GPUs, and the training converges in approximately 12 hours. The LoRA rank is set to 32 for all experiments. For a fair comparison with baseline methods, we use a video resolution of 480×832 during evaluation. We use Wan2.2-14B model as our backbone. We compare our method with CogVideoX-5B, HunyuanVideo-I2V-13B and WAN-2.2-14B baseline.

C. Comparison with Training on WISA

We compare our model, trained on only 1.5k synthetic videos, with a LoRA-based I2V model trained directly on the real WISA dataset [3] to evaluate synthetic-to-real generalization. As shown in Tab. 2 and Tab. 3, despite using *10× less* training data, our model achieves superior performance on both PIS and VBench I2V [1, 2]. These findings suggest that a compact synthetic dataset with accurate geometric and physical annotations can substantially reduce training cost while still improving physics-aware video generation quality.

From the comparison, we argue that the syn2real generalization is feasible because state-of-the-art pretrained video generation models already encode strong priors about visual dynamics. Our synthetic physic-clean data, enriched with explicit physics-aware annotations, acts as a targeted supervisory signal that elicits and reinforces these latent physical priors, enabling the model to better understand and utilize physics-based control cues. Our method can be trained on only 4 H100 GPUs in less than one day, offering an efficient and practical pathway for improving the physics-following ability of video generation models.

D. Additional Visualization Results

We provide additional video visualizations in a separate HTML file included in the supplementary material, including comparisons against baseline methods. We strongly encourage reviewers to view these results for a more comprehensive understanding of our model's physics-aware generation quality.

References

[1] Ziqi Huang, Yinan He, Jiashuo Yu, Fan Zhang, Chenyang Si, Yuming Jiang, Yuanhan Zhang, Tianxing Wu, Qingyang Jin, Nattapol Chanpaisit, Yaohui Wang, Xinyuan Chen, Limin Wang, Dahua Lin, Yu Qiao, and Ziwei Liu. VBench: Comprehensive benchmark suite for video generative models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024. 1

[2] Ziqi Huang, Fan Zhang, Xiaojie Xu, Yinan He, Jiashuo Yu, Ziyue Dong, Qianli Ma, Nattapol Chanpaisit, Chenyang Si, Yuming Jiang, Yaohui Wang, Xinyuan Chen, Ying-Cong Chen, Limin Wang, Dahua Lin, Yu Qiao, and Ziwei Liu. VBench++: Comprehensive and versatile benchmark suite for video generative models. *arXiv preprint arXiv:2411.13503*, 2024. 1

Table 2. Additional ablation study on VBench-I2V evaluation on the WISA-test set. Higher is better.

Method	I2V-Metrics		Quality Metrics					
	i2v_subject	i2v_background	subj_consis.	bkg_consis.	motion_smooth.	dynamic_degree	aesthetic	imaging
Train on WISA	0.965	0.971	0.927	0.944	0.991	0.612	0.452	0.627
Ours	**0.972**	**0.978**	**0.928**	**0.944**	**0.993**	**0.617**	**0.455**	**0.633**

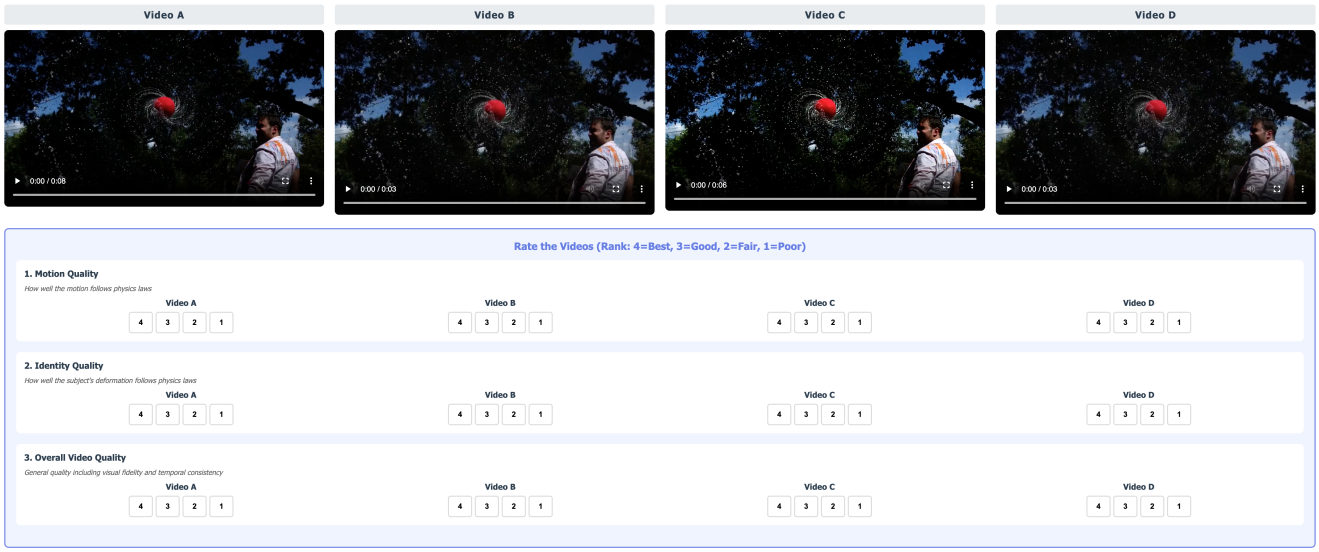

Figure 1. The visualization of our user study page.

[3] Jing Wang, Ao Ma, Ke Cao, Jun Zheng, Zhanjie Zhang, Jiasong Feng, Shanyuan Liu, Yuhang Ma, Bo Cheng, Dawei Leng, Yuhui Yin, and Xiaodan Liang. WISA: World simulator assistant for physics-aware text-to-video generation. *arXiv preprint arXiv:2502.08153*, 2025. 1